

Shammas Ahmed

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EDUCATION

Rice University

Houston, TX

B.S. Operations Research (specialization in Algorithms)

May 2027

GPA: 3.91/4.0

Relevant Coursework

Linear and Integer Programming, Convex Optimization, Discrete Optimization, Large-Scale Optimization
Stochastic Optimization, Online Optimization, Machine Learning with Graphs, Algorithm Design

EXPERIENCE

Undergraduate Research Assistant - Mixed Integer Programming

May 2025 – Present

Operations Research Department, Rice University

Houston, TX

- Investigating a **Path Planning Problem** with obstacle evasion and shortest paths with homology constraints
- Reformulated as a flow problem & identified the problem as NP-Hard
- Gained experience in convex optimization, flows, shortest paths, and SVD decomposition of complex matrices
- Actively working on a technical report for the problem

Undergraduate Research Assistant - Inverse Optimization

May 2024 – December 2024

Operations Research Department, Rice University

Houston, TX

- Conducted in-depth, novel research on **Inverse Integer Knapsack Problem** under faculty guidance
- Found geometric substructures to substantially reduce total computations (from *several millions* to *several tens*), plotted results in Python
- Short-listed as 1 of 50 participants at **Rice Summer Undergraduate Research Fellowship**

Teaching Assistant

August 2024 – Present

Rice University School of Engineering and Computing, Rice University

Houston, TX

- TA for Introduction to Engineering Computation (CMOR 220)
- Deliver weekly 1-hour lectures on concepts in math and engineering (e.g. neural networks, first-order optimization methods, root-finding methods) with implementation in MATLAB and Python
- Lead weekly office hours, grade assignments, provide feedback for 20+ students

PROJECTS

Homologous Path Planner | Python, Gurobi

May 2025 – Present

- Designed optimization-based path planner that finds the **shortest path for robots** under homology constraints
- Transformed paths into flows, implemented harmonic projection constraints on 20 x 20 grids
- Visualized path solutions in Python, built model using Gurobi, verified fast running times

Owl Help You Plan | Python, OR-TOOLS, Gurobi

February 2024 – April 2024

- Developed back-end mathematical model for **vacation itinerary optimizer**
- Applied LP, understood objectives and constraints, reduced problem to a *Vehicle Routing Problem*
- Finds best pathway to take around a city, visiting all *PoIs* and minimizing travel cost and time spent
- Conducted market research to create service without precedent, pitched at end-of-semester ceremony, distributed among 10-15 friends and family, used in practice on a trip with friends

POSTER PRESENTATIONS, CONFERENCES, SYMPOSIUMS

The Inverse of the Integer Knapsack Problem - Rice Summer Undergraduate Research Symposium

Aug 2024

The Inverse of the Integer Knapsack Problem - UT Arlington Math Symposium

Nov 2024

SKILLS

Programming Languages: Gurobi, Python, MATLAB, R, Java

Developer Tools: GitHub, Google Colab, VS Code, Visual Studio, PyCharm, IntelliJ, Excel

Research Techniques: LaTeX, Microsoft Office, poster making, poster presentation, data visualization, data analysis